COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
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NAME:	Kherwin Emmanuel D. Milan	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

Assignment 6.1 Linear and Binary Search

1. PROCEDURE OUTPUT

What procedures did you perform?

Answer the following questions:

1. What is a search tree in data structures?
   - A search tree is a data structure used to store and organize data in a way that makes searching for a specific value easier. It arranges the data according to certain rules so that we can find, insert, or remove elements more efficiently.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.
   - The two common search algorithms are linear search and binary search. Linear search checks each element one by one from the beginning until the desired value is found or the end of the list is reached. Meanwhile, binary search repeatedly divides a sorted list into two parts and checks the middle element. It removes half of the remaining elements after every comparison, making it much faster than linear search for large, sorted data.

3. What operations / implementations can be performed using binary and linear search operations?
   - Both binary and linear search can be used to find or locate a specific element in a collection of data. They can also be used to check whether a value exists in an array or list.

4. What are the advantages in using binary search tree as data structure?
   - A binary search tree can make searching, inserting, and deleting data faster when the tree is properly balanced.

5. Give an example program using binary search and Linear search.
   Linear

```
#include <iostream>
6.
7.  int main() {
8.      int arr[] = {10, 25, 30, 45, 50};
9.      int size = 5;
10.     int target = 30;
11.     int position = -1;
12.
13.     for (int i = 0; i < size; i++) {
```

```
            if (arr[i] == target) {
                position = i;
                break;
            }
        }

        if (position != -1)
            std::cout << "Element found at index " << position;
        else
            std::cout << "Element not found";

        return 0;
}
```

```
Element found at index 2
```

Binary

```
#include <iostream>

int main() {
    int numbers[] = {10, 20, 30, 40, 50};
    int size = 5;
    int search;

    std::cout << "Enter number to search: ";
    std::cin >> search;

    int left = 0;
    int right = size - 1;
    int position = -1;

    while (left <= right) {
        int middle = (left + right) / 2;

        if (numbers[middle] == search) {
            position = middle;
```

```
            break;
        }
        else if (numbers[middle] < search) {
            left = middle + 1;
        }
        else {
            right = middle - 1;
        }
    }

    if (position != -1) {
        std::cout << "Number found at position " << position + 1 <<
std::endl;
    } else {
        std::cout << "Number not found" << std::endl;
    }

    return 0;
}
```

```
Enter number to search: 40
Number found at position 4
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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